Bharath K. Pillai

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Education

The Ohio State University — Columbus, OH
B.S. in Computer Science & Engineering | Minor: Robotics and Autonomous Systems — *Expected May 2028*

Skills

Programming Languages: Python, Java, Rust, C, Assembly
Frameworks: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, Matplotlib
Developer Tools: Git, Docker, AWS, Raspberry Pi, ESP 32, VS Code
Relevant Coursework: Data Structures & Algorithms, Intro to Database Systems, Electronics for CS, Linear Algebra

Experience

National Science Foundation ICICLE AI Institute — Columbus, OH
Undergraduate Researcher – Computer Vision & Edge Systems — *May 2025 – Present*

- Built a model to get detected animals GPS coordinates using HFOV, camera bearing, and Vincenty geodesic math
- Validated GPS localization results against acoustic data, demonstrating cross-modal agreement with ∼80% overlap
- Merged multi modal audio and camera data using Pandas, NumPy, and OpenCV reducing reporting time by ∼70%

Ohio State University Dept. of Engineering & National Security Agency — Columbus, OH
Undergraduate Researcher – Software Defined Radio Systems — *Mar 2026 – May 2026*

- Implemented a SPECK128/256 cipher for an anti-jam frequency algorithm, tested against NSA vectors
- Developed diagrams for frequency assignment and bitsliced index computation for multi-user channel hopping

OnRamp Hub: Ohio | Defense Innovation Unit (DIU) Program — Columbus, OH
Subject Matter Expert Intern — *May 2026 – Present*

- Evaluating early-stage technologies from vendors, creating technical capability summaries for DoW partners
- Meeting and researching emerging critical tech, maintaining a database of companies and R&D trends

Projects

Wildlife Tracking Drone | *Python, YOLOv8, TensorRT, DeepSORT, MAVLink, RealSense* — Present

- Designing and deploying an autonomous drone that tracks wildlife using an onboard Jetson Nano
- Using YOLOv8 + TensorRT at 60 FPS, DeepSORT multi-target tracking, monocular depth via Intel RealSense D435i, and optical flow for velocity estimation
- Perception feeds a PID control loop generating commands to a Pixhawk 6C over UART in GUIDED mode

BioHack 2nd Place | *Python, PyTorch, OpenCV, Blender* — Feb 2026

- Created a deep learning workflow into vector embeddings for analysis across heterogeneous kidney imaging
- Preprocessed medical imagery with OpenCV by segmentation and masking for clear inputs for model inference
- Modeled kidney cortex using BPY in Blender, creating NURBS curve tubules with procedural sinusoidal winding
- Trained a self-supervised autoencoder in PyTorch to compress 512-d embeddings to 64-d latent space

MakeIO 3rd Place | *Python, YOLOv8, TensorFlow, PyTorch, Raspberry Pi* — Mar 2026

- Built a multimodal edge AI pipeline combining YOLOv8 detection with an audio CNN for real-time wildlife health
- Achieved 0% accuracy degradation vs. supercomputer server baseline at 14.9x lower energy consumption
- Designed a CNN processing 24 features achieving a 36% runtime reduction on ARM vs. standard architecture

Publications

Cross-Modal Corroboration for Annotation-Free Wildlife Monitoring — Jun 2026
First Author – CV4Animals Workshop, CVPR 2026

- Developed a self-validating framework combining BioCLIP 2 vision and acoustic classification, cross-checked against behavioral ecology priors to eliminate manual annotation, validated on a Milu deer herd

SmartWilds: Multimodal Wildlife Monitoring Dataset — Sep 2025
Co-Author – Imageomics Workshop, NeurIPS 2025

- Contributed to a peer-reviewed dataset integrating drone imagery, camera traps, and bioacoustics for conservation-scale computer vision, species detection, and habitat monitoring research